# Building a unified science of sensorimotor control with Densely Observable Measurment Environments [DOMEs]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

## 1. Mission

> We will create a new scientific observable representing the closed loop of the **perceptual, mechanical, and environmental** aspects of goal-directed behavior of freely moving organisms to reshape perceptual/motor neuroscience, muscuskeletal biomechanics, embodied/agentic artificial intelligence, and mobile robotics into a new convergent science of **sensorimotor control.**

### 1.1. Buiding the Dome

We will develop, validate, and broadly disseminate a **novel research instrument** we call a **Dome** *(Densely Overlapping Measurement Environment, Ref - Figure 1 - Hero Figure showing full DOME)*, which unifies a heterogeneous array of multi-modal sensors into single, centrally-controllable, metrologically-validated instrument for the measurement and manipulation of a freely moving agent within a specified 3D volume of real-world space (ref DOME-L, DOME-S, and DOME-Mobile in `3. Outcomes`).

We will organize our technical development around the technical bottlenecks which limit our ability to combine reconstructions the **eye**, **body**, and **environment** into combined represetations whch trace to activity in the central and peripheral nervous system, tagetting development effort on the target of tracking the body, eye, and world with sufficient precision and accuracy so that the retina-projected real-world stimulus is reliable enough to be targetted by cortical measurements in animal models, and body kinematics and kinetics are reliable enough so that surface and implanted EMG measures match full-body muscuskeletal activation models (e.g. via OpenSim or similar).

By metrologically tracing masuremnets from the heterogeneous sensors along both the Visual and Motor paths across a core set of cross-validating DOME variants built in the FMC-X flagship facility and fanned out to a network of cross-disciplinary collaborators to

- Internally consistent and metrologically-grounded data set is defined for auto-improving training (e.g. reprojection error to improve tracking models),
- backfill prior unmeasured DoFs with models trained on correlations in new DOME measurements that specify that DoF.
- Upfill data from lower-quality DOMEs (e.g. classrooms) based on models trained on flagship data.
- Unify research from disparate field through use-directed partially hydrating DOMEs based on research domain.
- Train next gen of embodied autonomous data from densely overlapping data and metrologically grounded downstream estiamtes i.e. Inverse Reinforcement Learning on human/animal data to infer viable strategies, RL models to guide robots use to design controlled experiemnts on humans)

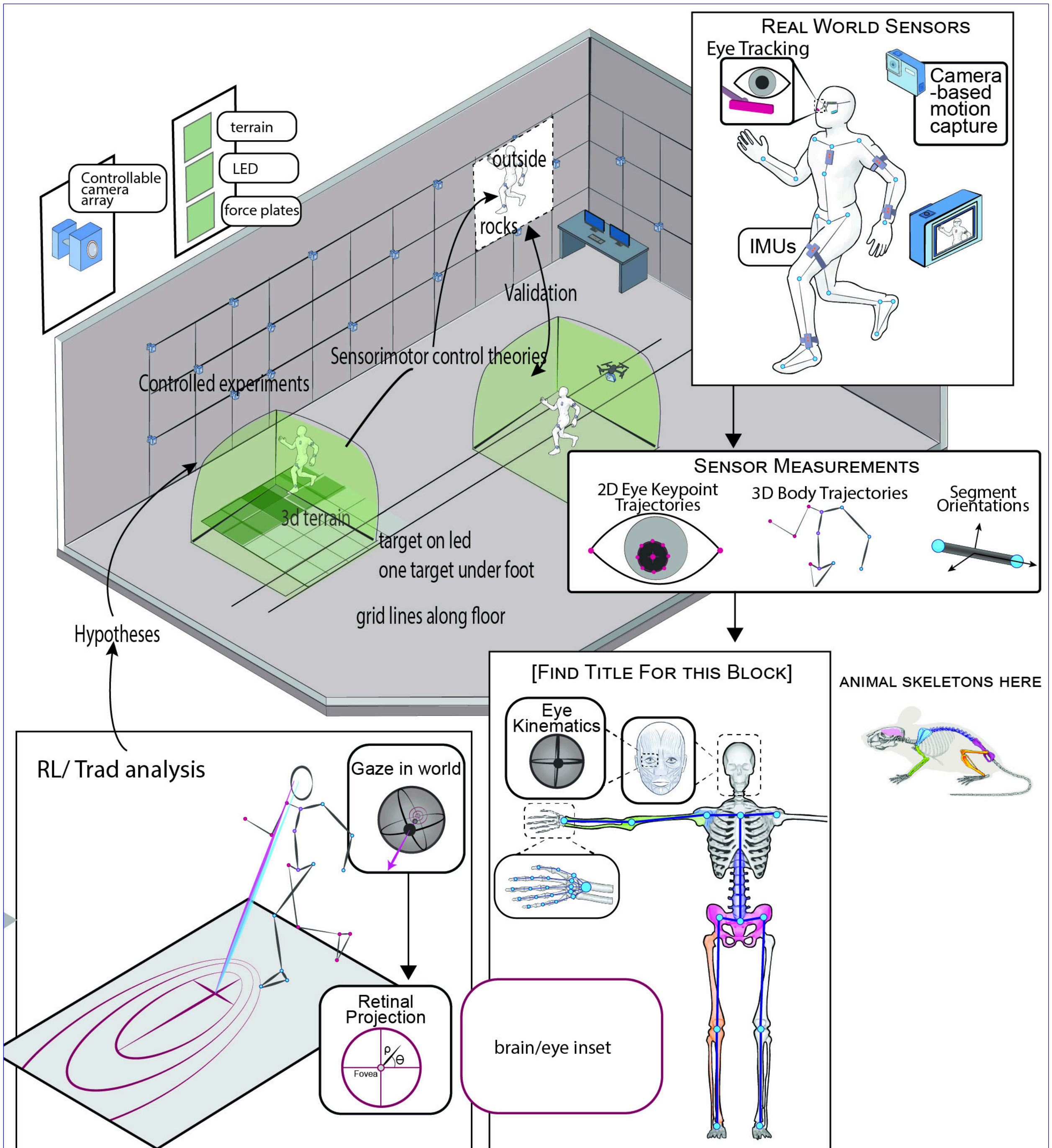
### 1.2. Unmet instutional needs

- Conway's Law
  - Academia too siloed for true cross-disciplinary works

  - Funding mechanisms defined as single-discpline institutions. Impossible to properly target proposals to announcments
  - Rome-on-top-of-Rome development of academic tools defines complexity ceiling for research tools (SKP-EI to help build to global scale)
- System rewards novel discoveries, not long slow work of proper tool making (Tools should be built by masters and taught to students, not other way around)

#### 1.2.1. Vision

- We define ourselves by the tool and the measurement, drive our development by direct needs of collaborator network.
- Form the boundary object between fields
- DOME - makes
- Best Eye trackers ever made at any price
- Best Mocap system ever made at any price
  - (but we'll sell em **real** cheap)
- Completely new fully integrated platform, exposing entirely new observable defined by densely overlapping measurements from the DOME
- DOMEs **everywhere**
  - Every human or animal-focused lab
  - Every hospital and PT clinic
  - DOMES on the space station and Vomit Comet rocket (study sensorimotor control and VOR in microgravity)
  - DOMES in athletic centers - Training athletes for high performance through education into their underlying sensorimotor control systems
  - DOMES in mobile robot and drone labs - Align with human research, stop the robot folks from handrolling their own mocap all the time
  - DOMES at the bottom of swimming pools and integrated into SCUBA systems of underwater welders (and other specialized trades)
  - Mobile DOMES studying animals in natural enviornment, mounted to the bottoms of ships and integrated with ultrsonic sensors to mocap dolphins playing in the wake
  - Tiny DOMEs to study honeybee and dragonfly visual flight control
  - DOMES in every classroom!
    - In both science/tech/health classrooms AND animation/game design classrooms (DOMEs for animation mocap, freemocap already proved this works)
    - Always build "cheaper than a textbook" version of all hardware
    - Comensurate data output (with less precision and fewer modalities)
    - Prepare new generation for a future where human and robot centered research is focused on full-body kinematics and 3d binocular eye tracking with retinal projection.
    -

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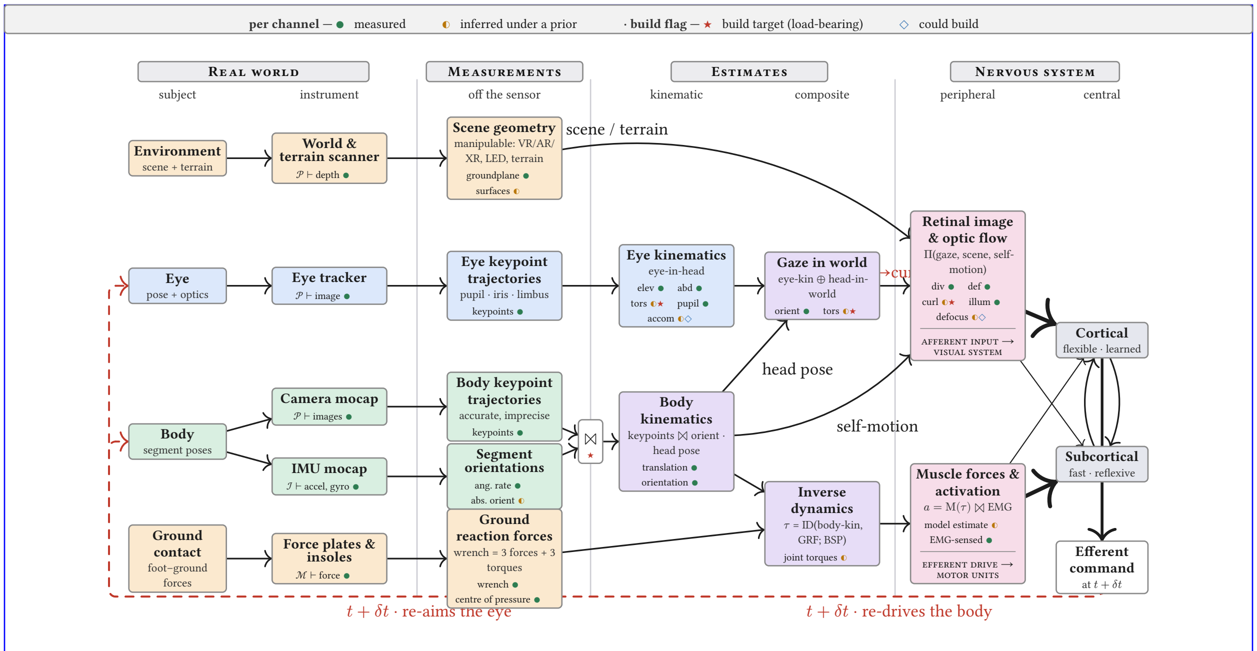


previous instant $\longleftarrow$ $t - \delta t$ · **this figure — a single instant $t$** · $t + \delta t$ $\longrightarrow$ next instant *the efferent loop advances the frame by $\delta t$*

**READING THE CHART AS A CALCULUS** each value is a bundle of **channels** (degrees of freedom); the top legend grades each channel.

**Notation**

| | |
|---|---|
| $\mathcal{E}$ | an **energy** a sensor is sensitive to: $\mathcal{P}$ photons · $\mathcal{J}$ inertial · $\mathcal{M}$ contact |
| $\mathcal{E} \vdash s$ | **transduce** — a sensor **yields** ($\vdash$) a raw **signal** $s$ from an energy (image · inertial · force) |
| $\rightarrow$ | **derive** — a signal is processed into a measurement, then an estimate |
| $\oplus$ | **compose** — chain relative poses; the shared reference frame cancels |
| $\bowtie$ | **fuse** — union the measured subspaces of two estimators |
| $\Pi$ | **project** — the only operator that maps input DOF onto output channels |
| $\approx v^\star$ | **approximates** the true latent value $v^\star$ |

**The chains** — energy $\vdash$ signal $\rightarrow$ measurement $\rightarrow$ estimate

| | |
|---|---|
| **Eye** | $\mathcal{P} \vdash$ image $\rightarrow$ eye keypoints $\rightarrow$ eye-kin (elev, abd, tors, pupil, accom) |
| **Body** | $\mathcal{P} \vdash$ images $\rightarrow$ body keypoints; $\mathcal{J} \vdash$ accel+gyro $\rightarrow$ segment orient; body-kin $=$ keypoints $\bowtie$ orient |
| **Gaze** | $g =$ eye-kin $\oplus$ head-in-world |

**Into the nervous system**

| | |
|---|---|
| **Retinal input** | $\rho = \Pi(g, \text{scene}, \dot{x}) \approx \rho^\star$ retinal image & optic flow (afferent) |
| **Kinetics** | $\mathcal{M} \vdash$ force $\rightarrow$ GRF; $\tau =$ ID(body-kin, GRF; BSP) |
| **Motor** | $a = \mathrm{M}(\tau) \bowtie \mathrm{EMG} \approx a^\star$ muscle drive (efferent) |

Figure 2: **The measurement chain of a DOME.** Reading across four bands — **real world ⟶ measurements ⟶ estimates ⟶ nervous system.** Sensors transduce physical energies into measurements (one step off the sensor); fusion ($\bowtie$) and pose composition ($\oplus$) build the kinematic and composite estimates that project ($\Pi$) onto the two points where the body meets the nervous system — the retinal stimulus entering vision and the muscle drive leaving the motor system. Every box is a bundle of **channels**; each channel is marked on two axes — measured (●) or inferred under a prior (◐) now, and, if unmeasured, a committed build target (★) or merely feasible (◇). The eye path makes the argument concrete: ocular torsion is inferred under Listing's law today and corrupts exactly the retinal **curl** channel — which is why it is a build target. The two peripheral signals project to the **central nervous system** (cortical + subcortical, reciprocally coupled), whose efferent command re-aims the eye and re-drives the body at the next instant — closing the loop. The whole figure is a **single instant $t$**: it is preceded by the infinitesimal before it and followed by the one after, and the dashed red loop is what advances the frame by $\delta t$. The calculus strip beneath the diagram gives the full reading rule, every symbol defined.

## 2. Technology Landscape

- FreeMoCap and its landscape
  - Established Tool with Growing International Community
    - >15k uses from 152 countries, 10k GH stars, 3.5k in Discord server
  - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
  - v2 has realtime and professional architecture (thanks to EI)
  - DOME-L and DOME-S camera-based mocap will use FMC or a variant. Mocap, Eye, World camera software composable from FMC polyrepo (skellycam, skellytracker, skellyforge, etc)
- ARGP
  - ARGPv1 - PhD Era[1–4]
  - ARGPv2 - NEU Era (Fuzzed out due to complexity, frustration spawned freemocap)
  - DOME includes ARGPv3
- Laser Skeleton research
  - [5–7]
  - etc
  - Ferret work
    -
- Existing Instruments
  - **Modern mobile eye trackers suck**
    - Pupils labs - Used to be cool, black box proprietary now. Best Available, used in Ferret and Laser Skeleton work.
    - Tobii et al not worth mentioning
    - Missing DoF's!
      - Can't measure Torsion
        - Field has invoked Listing's Law to ignore torsion for decades
        - Listing's Law is NOT APPLICABLE when VOR is active
        - VOR is active during 100% of natural behavior
        - Torsion directly affects axis that Retinal Curl[6] lives on, so inabilty o measure it corrupts signal we want to propogate into visual cortex
        - We know how to measure Torsion (iris texture), and Lens Accomdation too, while we're at it (yates et al DPI)
  -
  - **Modern mocap sucks**
    - Old, awful, expensive, black boxes
    - So difficult to use its a literale bottlenecks
    - Difficult limits lab size (Cant re-target cameras without shutting down the lab for a week. Can't maintain multiple configurations)
    - Theia exists, but its re-building the same problems of the previous generation in a new modality
  - **CV/ML tools exist and are cool, but all single threads, no unified tool**
    - OpenSim - Widely used, very difficult to operate, highly specialized niche within biomechanics, single-modality
    - DeepLabCut - Widely used, very difficult to operate, single-modality

  - OpenCap works, but only for iPhones and requires a cloud upload, and caters exclusively to clinical biomechanics. Single-modality, non-convivial.

- **CV is drifting away from emprically grounded tools**
  - Progress on the empirically useful parts of the CV stack (e.g. pose estimation, optical flow) hasn't stopped, but its def stalled ("too close to the pixels" - more focus on "sematic segmentation" and single-camera scene inference - cool and useful, but not empirically grounded )
  - Things still reference benchmarks on increasingly old COCO data scaped from early 2000′s flikr data, limitations havent improved. Gaps for studying patient populations and exteme human movement (acrobatics, circus, etc)
  - **SIMPL/Meshcapade** - Good example of building past empirically validity. 3d from 2d and mesh-fitting pipelines not truth preserving. Building increasingly more abstract and less tracable data channels on top of the same 2d data we've had for a while
  - Folks like **R James Cotton** doing great work, but single application exploration cannot scale beyond fairly nice application within clinical biomechanics

- **Mobile Legged Robot revolution**
  - Unbelievable progress in legged robots in recent years
  - Largely atributable to advances in RL from physical simulations ( e.g. MuJoCo/Nvidia IsaacLab ) - [*Note that that is MY read on the recent progress of robots, we should try to back it up somehow or just state it as fact and move on? I'm pretty confident that its true enough for govt work, as they say*]
  - We can feed them directly from our DOME, make it easy for ANY robot lab to build their own DOME. Build DOMEs in previous hard to record settings (e.g. deep sea welding) to develop control theory for next gen autonomous robots
  - Control theory based on RL for Robots -> Drive controlled experiments on humans and animals
  - Inverse Reinforcement Learning on human/animal agents, jump start RL training runs for task-matched autonomous agents.

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## 3. Outcomes

- Develop, validte, and disseminte - DOME Variants
  - **DOME-L** — the flagship instrumented volume, in the greater Boston area. Large enough to fully enclose the smaller variants for validation and metrological grounding.
  - **DOME-S** — the extension of the commodity-camera capture volumes already built and disseminated through the FreeMoCap Project. Representative of the DOME a research lab or a classroom would build.
  - **DOME-Mobile** — a wearable, self-contained DOME producing Dense Observations in unconstrained indoor and outdoor environments, continuing the lineage of the PI's gaze/gait and retinal optic flow work[5–8].
    - Validate in large DOME-L so we can trust the outdoors
  - Also - disseminate custom use-directed builds of DOME variants through network of collaborators studying Humans, animals, and robots.

### 3.1. Bottlenecks

- **Technology**
  - **New eye tracker**
    - Best ever made
    - Add every sensor we can, see which allow us to measure DoFs we need, optionally scale back and remove
    - Optionally also make cheaper version for classroms - fully compatible data output (with less precision and depth)
      - Remake the field by retraining the next generation of researcher in the tools of the field you want to build
    - Dense instrumentation in the eye and world cameras
      - **Eye sensor array**
        - includes
          - High res IRGB
          - High speed IRGB
          - MEMS single-pixel sensor
          - LIDar scanner
          - IR/Structured IR illuminators
          - Event Camera
      - **World sensor array**
        - Build differnt form factors (big heavy, small light)
        - Head- or Drone- mountable
        - Stereo RGB, Structured IR, LiDAR, etc
        - High density multi-model environment scan
        - Fast scan (RGB-D)
        - Slow scan (Textured Mesh)
        - Semantic segmentation (via SAM3, YOLO, etc)
        -
- **Auto-configuring Mocap Camera Array**
  - Large scale mocap too hard to control, cuts against usefulness of large volumes. Legit barrier to progress and usefulness of mocap!!

  - Our Auto-calibrating, Programmtically controlled Mocap Camera array will help!
    - Allow centrally controlled definition of capture volume within availble range
    - Auto-configure for high density tracking within a small space, or more diffuse tracking across a larger space. Higher ceiling for drones, or groundplane focus for locomotion
    - Use to cross-validate smaller DOMEs within its space, Programmtically explore best practices for camera placement and calibration procedures
  - **Each camera**:
    - Fully actuated Extrinsics (pitch, tilt, roll motor) and Intrinsics (Zoom, apperture, focus)
    - **Power Over Ethernet** (daisy-chainable power + data, so we can control arbitrary cameras from a single wire coming off the computer)
    - Modular:
      - Sensor (high def, high speed, spectral sensitivity, size, etc)
      - Illumination (Matchin spectrum of sensor)
      - Lens Assembly (zoom, aperture, etc)
- **Hybrid Body kinematics**
  - Fuse IMU- kinematics iwth Camera-kinematics for BEST kinematics (able to do inverse dynamics (note - also needs force plates))
- **Drone swarm mocap**
  - Do it in comparison of IMU- mocap to static DOME cameras, then itnegrate a **Swarm of Drone-mounted World-Sensor arrays** that ground inside-out estiamtes of IMU- and head-mounted World-sensor array during unconstrained outdoor used (train drones and develop swarm controller by tracking them within large DOME-L)
  - Drone kinematics will be noisy, but fusable with IMu-mocap for grouned outdoor data
- **Modular Force Plate floor panels**
  - Floor is all removable panels
  - Integrate force plates at key points
  - Create **Terrain** and **LED sceen** panels that can attached to top of force plates, give force reading under defined 3d terrain and interactive LED panels (for cntrolled experiments that manipulate location and texture motion of target footholds)
  - Each 0.5m square of the ground can be: Forceplate, Terrain/LED panel, or both (force plate stacked with terrain that is either inert or composed of LED panels)
  - Force plates allow measuring GRF and downstream OpenSim and Inverse Dynamics stuff to estimate muscle activity
  - Terrain allows validation of head and drone mounted world scanners
  - LED panels allow controlled experiments of visual/locomotor control theory, and useful human DOME data to use Inverse Reinforcement Learning to extract apparent human control policy, and feed that policy AND and the actual DOME data into robot RL control systems solvers

### 3.2. Timeline

NOTE - THIS IS A PLACEHOLDER!!!! THIS IS A PLACEHOLDER

| Phase | Name | Description |
|---|---|---|
| 1 | **Organize** | Set up Organization (Phase 0) — Solidify release of FreeMoCap v2 |
| 2 | **Architect** | Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems) |
| 3 | **Execute** | Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system |
| 4 | **Stabilize** | Establish OODA loop within expected standard operations |
| 5 | **Expand** | Once stabilized, expand key performance metrics |
| 6 | **Align** | Focus on building deeply integrated community of users and contributors (Transition BDFL ⟶ BDFaSPoT) |
| 7 | **Establish** | Focus on long term stability and healthy growth |
| 8 | **Maintain** | Beyond proposal timeline — Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending ‘mesoscale research revolution’ to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus) |

Table 1: Timeline of Phased Milestones

NOTE - THIS IS A PLACEHOLDER!!!! THIS IS A PLACEHOLDER

NOTE - Names abbreviated in public draft. Check initials to real names in local .env (or request it, if you need it)

| Name / Role | Qualifications |
|---|---|
| **Jonathan Matthis, PhD**<br>President | • Founder and chief maintainer of FreeMoCap project<br>• Left tenure track academic research position precisely because the institution could not support the work that needed to be done<br>• Decades experience integrating visual neuroscience with biomechanics/ robotics[5–7] |
| **NR**,<br>CEO | • Expertise in technically and software dev<br>• Previous CEO XP, founded and ran own buisiness<br>• CEO helps PI apply his specific expertise to the core tasks of the XLab |
| **EI**<br>CTO | • Decades of experience in software Industry<br>• Chief architect and CTO for established companies<br>• Represents expertise truly absent from academic contexts<br>• Ensures world-class enterprise-level software capability<br>• Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science |
| **AC, PhD**<br>CSO | • Expertise in validation of clinical research tools and bench-to-bedside development<br>• [CITE DISSERTATION]<br>• Experience in neuroprosthetics design and brain imaging<br>• Worked with PI on core FMC dev and validation |
| **JKL, PhD**<br>CAIO | • PhD - Cognitive Science<br>• Director - Cognitive Science Program<br>• Current - Senior AI Applications Engineer @ SOLID<br>• Expertise in use of AI in classroom, lab, and enterprise |
| **RR**<br>CFO | • Bookkeeping<br>• NSF Grant management<br>• Finances<br>• Entrepreneurship (founded own Bookkeeping coop)<br>• Taught Finance Theory at university level |
| **KM**<br>Project Manager (?) | • Key driver of[7]<br>• Worked with PI on core tech[6]<br>• Expertise in key technologies |
| **MN**<br>Project Manager (?) | • Expertise in clinical biomechanics,<br>• Medical systems design,<br>• Mechatronics for clinical tools,<br>• Multi-modal motion capture lab management |

Table 2: Senior/Key Personnel Qualifications

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# 4. Team Capabilities Statement

## 4.1. Senior Team Capabilities

- Jon - done it before in humans, currentludoing it in ferrets and mice. Left tenure track position to run FreeMoCap, XLab funding will accelerate that mission
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development
- NR - XP as CEO and working in tech industry. Hiring her as CEO allows Jon/PI to focus on applying their domain expertise on the core work of the lab
- JKL - AI expertise in teaching, research, and industry. Will build internal AI systems for core software, developer, user, and student assistance.
- AC - focusing clinical adoption and bench-to-bedside transitions. Wrote his dissertation on validation of freemocap. Has all skillsets needed for DOME-S
- MN - XP in mechatronics and large-scale lab building and management. Hase core skillset needed for DOME-L
- KM - Part of original laser skeleton team, has all necessary sub-skills to buld core instrumentation of Mobile DOME

## 4.2. Collaborators

| Initials | Model / Domain | Expertise |
|---|---|---|
| BS | Ferrets | Visual/Perceptual neuroscience, ephys |
| MD | Guinea Fowl | Musculoskeletal Biomechanics, muscle units |
| DF | Mice | System Biology |
| AH | Marmosets / NHP | Ephys |
| MH | Human | Vision and CPS, Natural Bx, VR/AR and eye tracking |
| KB | Human/NHP | Visual Neuroscience, Optometry |
| JY | Human/NHP | Computational neuroscience, Eye tracknig, Ephys |
| GD | HUman | Eye tracking, VR/AR, Perceptual/Motor control |
| Mayo Clinic | Human | Brain scans and implanted sensors |
| BD, GN/RAI, CH, JH, SH | Robotics, Prosteheics, Exoskeletons | Control theory |

Table 3: Collaborator Network

## 4.3. Governance

- FMC-X will exist as a independent project within FMC-F organization
- FMC-F will handle other work (e.g. FreeMoCap Project), but otherwise the missions almost fully overlap
- Nearly identical leasdership between FMC-F and FMC-X, with final say on all topics falling to the President (who is also the PI) ensures fast decision making (no higher management to talk to - PI is the top of the org chart)
- Collaborator network and user community consulted for design decisions, work targets, and roadmap
  - Handled via “Request for Comments” style annoucments (like Python’s PEP), so progress can’t be stalled by community problems.

- Annual Workshop/Hackathon/COnference/Congress to bring together collaborator network to discuss progress, tech targets, friction, etc.
  - Workshops to train students and out-of-domain experts
  - Hackathons to on-board developers and foster in-person connections between normally remote workers (important for FOSS)
  - Conferences to discuss progress and rsearch done with DOME tech
  - Congress to dicuss org status, plans, and future (annual State of the Skelly address)
  - Developer fund, Community Grants Program, etc to support FOSS network
    - Use Community Grants Program to to test differnt strategies for community engagement and consensus taking (Eurovision style voting, DAO methods, etc)

### 4.4. Autonomy

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.
  - This project will happen no matter what, but with XLabs funding, we could truly change the world

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